

Ecuaciones integro-diferenciales

$$F_a(t) = (P_a(t) - L \frac{dF_a(t)}{dt} - \frac{1}{C} \int [F_a(t) - F_r(t)] dt) \frac{1}{Z}$$

$$P_p(t) = RF_r(t)$$

$$F_r(t) = \left(\frac{1}{C} \int [F_a(t) - F_r(t)] dt \right) \frac{1}{R}$$

Transformada de Laplace

$$P_a(s) = ZF_a(s) + LsF_a(s) + \frac{F_a(s) - F_r(s)}{Cs}$$

$$P_p(s) = RF_r(s)$$

$$\frac{F_a(s) - F_r(s)}{Cs} = RF_r(s)$$

Procedimiento algebraico

$$\frac{F_a(s)}{Cs} = F_r(s) \left(R + \frac{1}{Cs} \right)$$

$$F_a(s) = F_r(s) Cs \left(\frac{RCs+1}{Cs} \right)$$

$$F_a(s) = F_r(s) (RCs + 1)$$

$$P_a(s) = F_a(s) \left(Z + Ls + \frac{1}{Cs} \right) - F_r(s) \left(\frac{1}{Cs} \right)$$

$$P_a(s) = F_r(s) \left[(RCs + 1) \left(Z + Ls + \frac{1}{Cs} \right) - \frac{1}{Cs} \right]$$

$$P_a(s) = F_r(s) \left[\frac{ZRC^2s^2 + ZCs + RLC^2s^3 + LCs^2 + RCs + 1}{Cs} - \frac{1}{Cs} \right]$$

$$P_a(s) = F_r(s) \left[\frac{Cs(ZRCs + Z + RLCs^2 + Ls + R)}{Cs} \right]$$

$$P_a(s) = F_r(s) (ZRCs + Z + RLCs^2 + Ls + R)$$

$$P_a(s) = F_r(s) (RLCs^2 + s(L + ZRC) + Z + R)$$

Función de transferencia

$$\frac{P_p(s)}{P_a(s)} = \frac{F_r(s)(R)}{F_r(s)(RLCs^2 + s(L + ZRC) + Z + R)}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{R}{(RLCs^2 + s(L + ZRC) + Z + R)}$$

Control

$$\lambda_1 = \frac{-(0.2+1E3*2E3*10E-6) + \sqrt{(0.2+1E3*2E3*10E-6)^2 - 4(2E3*0.2*10E-6)(1E3+2E3)}}{2(2E3*0.2*10E-6)} = -153.16$$

$$\lambda_2 = \frac{-(0.2+1E3*2E3*10E-6) - \sqrt{(0.2+1E3*2E3*10E-6)^2 - 4(2E3*0.2*10E-6)(1E3+2E3)}}{2(2E3*0.2*10E-6)} = -4896.84$$

Caso

$$\lambda_1 = \frac{-(0.3+3E3*2E3*10E-6) + \sqrt{(0.3+3E3*2E3*10E-6)^2 - 4(2E3*0.3*10E-6)(3E3+2E3)}}{2(2E3*0.2*10E-6)} = -83.6144$$

$$\lambda_2 = \frac{-(0.3+1E3*2E3*10E-6) - \sqrt{(0.2+1E3*2E3*10E-6)^2 - 4(2E3*0.2*10E-6)(1E3+2E3)}}{2(2E3*0.2*10E-6)} = -9966.3856$$

Error en estado estacionario:

$$E(s) = \lim_{s \rightarrow 0} \left[S * \frac{1}{s} \left(1 - \frac{R}{(RCL)s^2 + (ZR_2C + L)s + Z + R} \right) \right]$$

$$E(s) = 1 - \frac{R}{Z+R}$$

$$E(s) = \frac{Z+R-R}{Z+R}$$

$$E(s) = \frac{Z}{Z+R}$$

Control.

$$Z = 1k\Omega$$

$$R = 2k\Omega$$

$$E(s)_{control} = \frac{1k\Omega}{1k\Omega + 2k\Omega} = 0.33$$

Caso.

$$Z = 3k\Omega$$

$$R = 2k\Omega$$

$$E(s)_{caso} = \frac{3k\Omega}{3k\Omega + 2k\Omega} = 0.6$$